ERNIE: Enhanced Representation through Knowledge Integration

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Introduction

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Language representation pre-training has been shown effective for improving many natural language processing tasks (such as named entity recognition, sentiment analysis, and question answering)

Recently, lots of works such as Cove, Elmo, GPT, and BERT improved word representation via different strategies

The vast majority of these studies model the representations by predicting the missing word only through the contexts.

Introduction

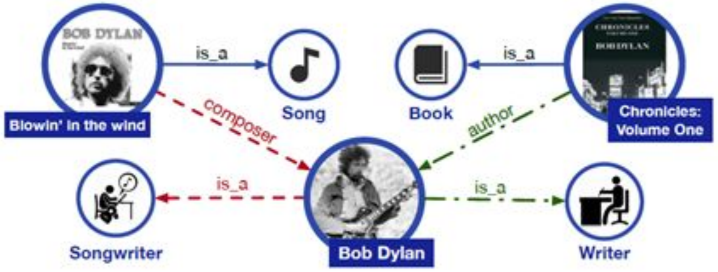
But, These works do not consider the prior knowledge in the sentence.

It is intuitive that if the model learns more about prior knowledge, the model can obtain more reliable language representation.

ERNIE (enhanced representation through knowledge integration) by using knowledge masking strategies.

In order to reduce the training cost of the model, ERNIE is pre-trained on heterogeneous Chinese data, and then applied to 5 Chinese NLP tasks.

기존 pre-trained LM의 한계



Bob Dylan wrote ***Blowin' in the Wind*** in 1962, and wrote ***Chronicles: Volume One*** in 2004.

- Though they have achieved promising results,
  they **neglect to incorporate knowledge information**
- 위의 그림에서 ***Blowin' in the wind***가 **음악**이고, ***Chronicles: Volume One*** 은 **책**이라는 사전지식이 없다면 ?
  → 주어진 문장 내 *Bob Dylan*의 직업이 가수이고 작가라고 판단하기 어려울 것
- 최근 관련 연구 결과에 따르면, 추가적인 지식 정보를 더해줌으로써 기존 모델의 성능을 유의미하게 향상시킬 수 있는 것으로 나타남

Knowledge information을 고려한 모델의 필요성 대두

How ERNIE tackles them?

- **Structured Knowledge Encoding**
  - first extract the entities in the text
  - then align the entities with the knowledge map
  - The knowledge map is first encoded by a knowledge embedding method (such as TransE) and then used as an input to the ERNIE model.
- **Heterogeneous Information Fusion**
  - utilizes a BERT-like architecture (adopting MLM & NSP as pretraining objectives),
  - adds a new pretraining objective for better named entity alignments.

Related Work

2.1 Context-independent Representation

Traditional methods focused on context-independent word embedding.

Methods such as Word2Vec (Mikolov et al., 2013) and Glove (Pennington et al., 2014) take a large corpus of text as inputs and produces a word vectors, typically in several hundred dimensions.

They generate a single word embedding representation for each word in the vocabulary.

However, a word can have completely different senses or meanings in the contexts.

2.2 Context-aware Representation

- **Skip-thought (Kiros et al., 2015)**
- proposed a approach for unsupervised learning of a generic, distributed sentence encoder.

- **ULMFit (Howard and Ruder, 2018**
    - effective transfer learning method

2.2 Context-aware Representation

- **ELMo (Peters et al., 2018)**
  - Word embedding research along a different dimension
  - to extract context-sensitive features

- **GPT (Radford et al.,2018)**
  - to enhance context-sensitive embedding by adapting the **Transformer**

2.2 Context-aware Representation

- **BERT (Devlin et al., 2018)**
  - two different pretraining tasks
    - randomly masks a certain percentage of words in the sentences
    - learn to predict those masked words
    - predict whether two sentences are adjacent.

- **MT-DNN (Liu et al., 2019)**
  - task information into the pre-training process and adapt their model to zero-shot tasks

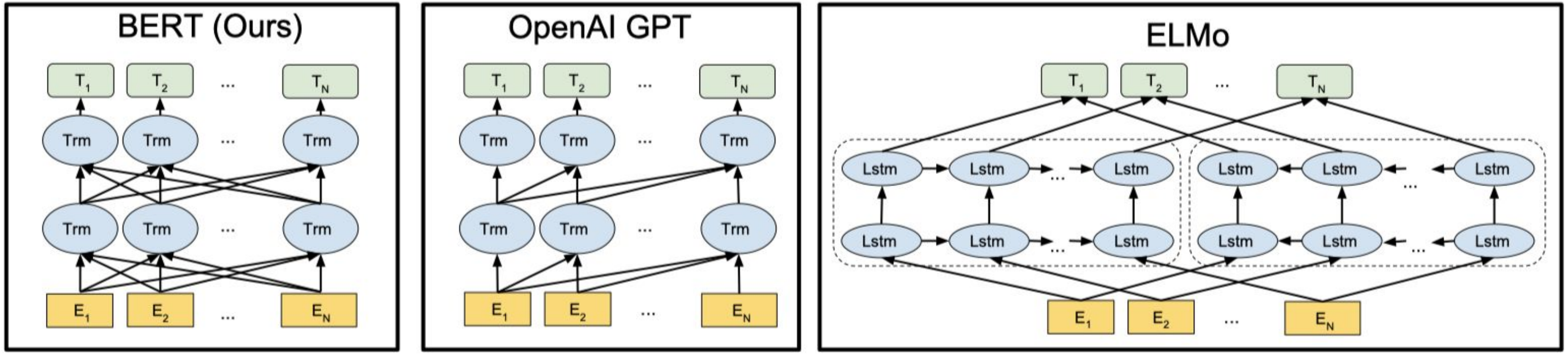
Differences (BERT, OpenAI GPT, ELMo)

- feature-based approach: the sequence-level model(ELMo 2018)
- Fine-tuning approach: OpenAI GPT, BERT (2018)

- Bert uses a bidirectional Transformer

- OpenAI GPT Uses a left-to-right Transformer

- ELMo uses the concatenation of independently trained left-to-right and right-to-left LSTMs to generate features for downstream tasks



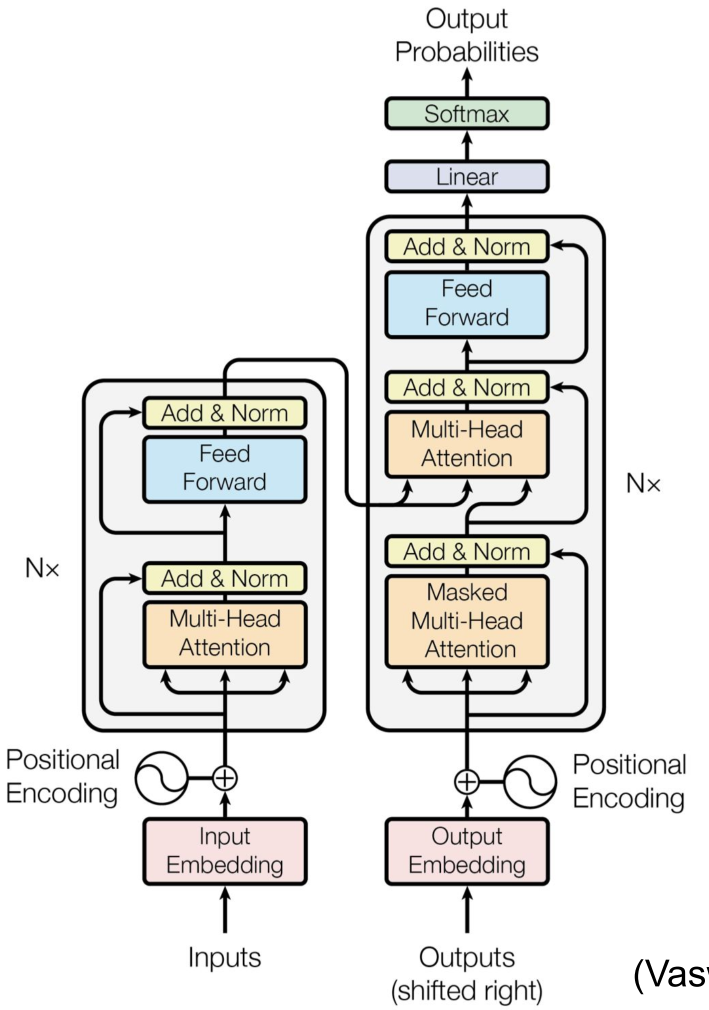
2.3 Heterogeneous Data

- **Universal sentence encoder (Cer et al., 2018)**
  - training data
    - Wikipedia, web news, web QA pages, discussion forum
- **Sentence encoder (Yang et al., 2018)**
  - based on response prediction benefits from query-response pair data drawn from Reddit conversation
- **XLM (Lample and Conneau, 2019)**
  - Parallel corpus to BERT
  - With **transformer model** pre-trained on heterogeneous data

Methods

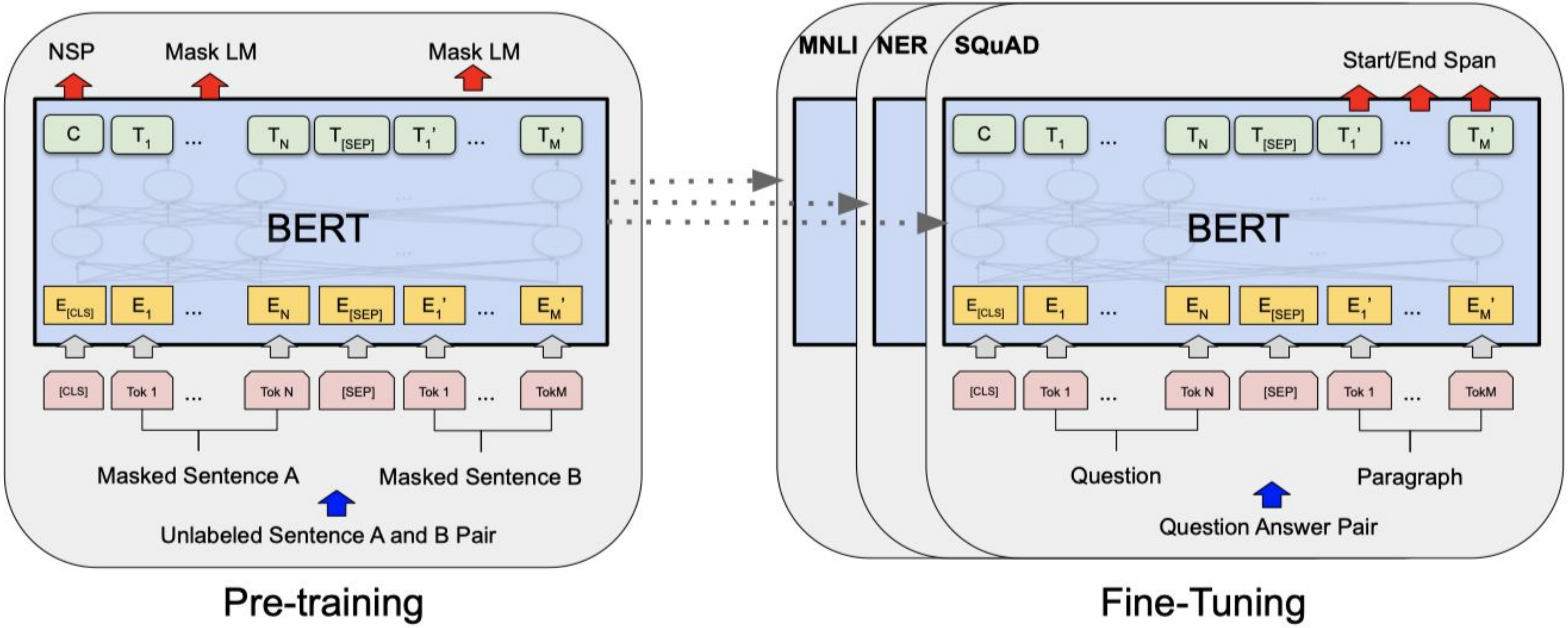
Transformer

Model architecture

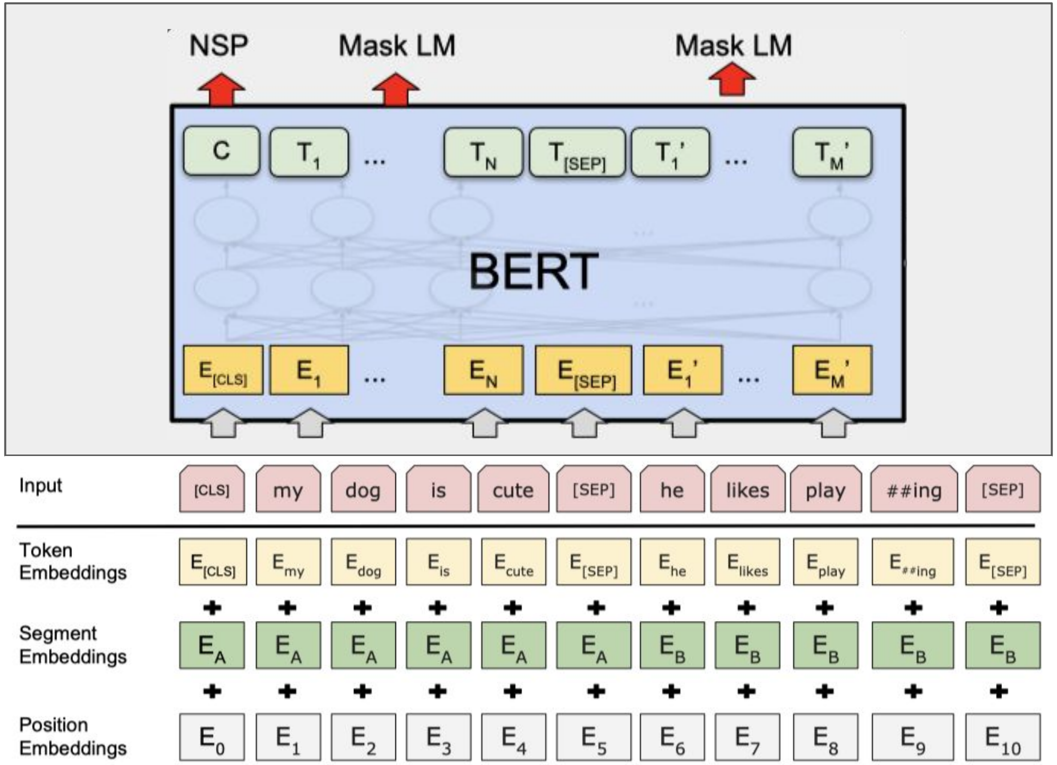


(Vaswani et al. 2017)

Pre-training & fine-tuning procedures for BERT

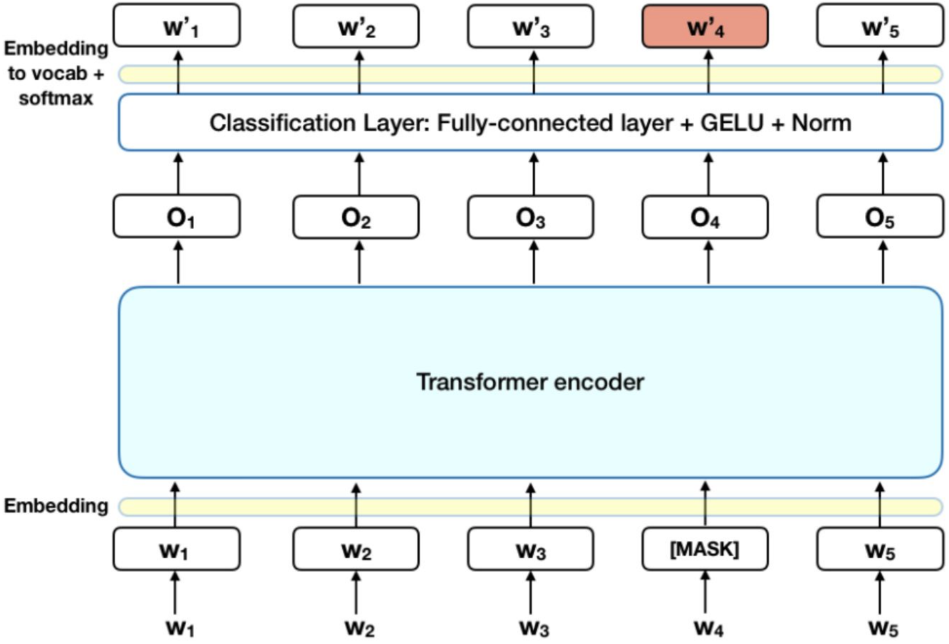


Input/Output Representations



Pre-training BERT

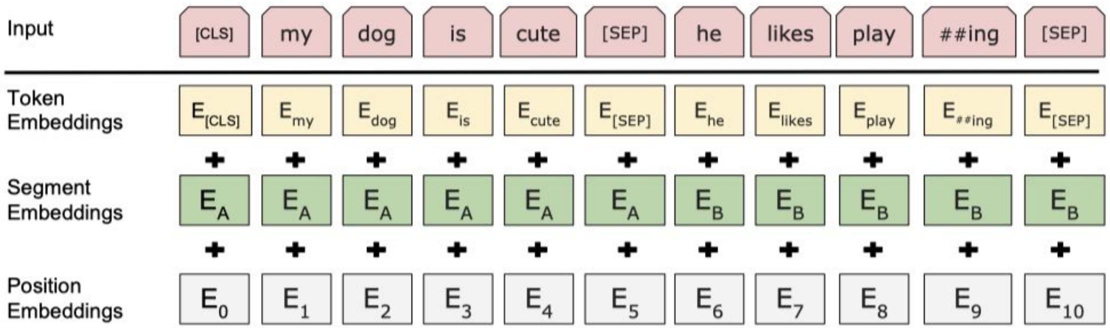
Task #1: Masked LM



- 80% of the time: Replace the word with the `[MASK]` token, e.g., `my dog is hairy` → `my dog is [MASK]`
- 10% of the time: Replace the word with a random word, e.g., `my dog is hairy` → `my dog is apple`
- 10% of the time: Keep the word unchanged, e.g., `my dog is hairy` → `my dog is hairy`. The purpose of this is to bias the representation towards the actual observed word.

Pre-training BERT

Task #2: Next Sentence Prediction (NSP)



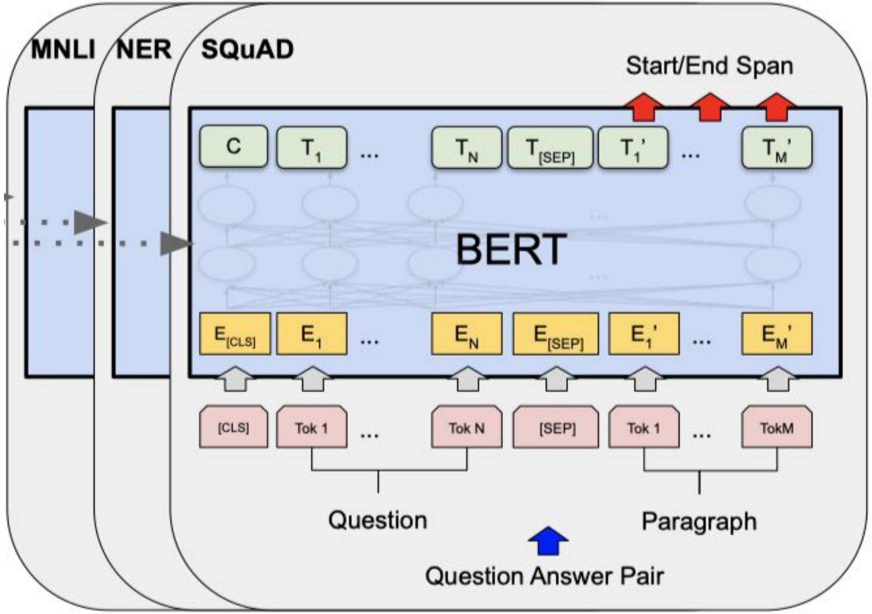
Input = `[CLS] the man went to [MASK] store [SEP]`
`he bought a gallon [MASK] milk [SEP]`

Label = `IsNext`

Input = `[CLS] the man [MASK] to the store [SEP]`
`penguin [MASK] are flight ##less birds [SEP]`

Label = `NotNext`

Fine-tuning BERT



Fine-Tuning

- **Batch size**: 16, 32
- **Learning rate (Adam)**: 5e-5, 3e-5, 2e-5
- **Number of epochs**: 2, 3, 4

3.method

BERT에서 language feature를 적용해서 masking함

- Multi layer transformer encoder 사용 : Bert와 동일
- Knowledge Transfer : language knowledge 사용
    - Basic level masking: bert와 동일 15% random masking
    - Phrase level masking: lexical analysis를 통해 phrase를 추출하고 masking함
    - Entry level masking: NER를 통해 entity 추출 후 masking

Sentence	Harry	Potter	is	a	series	of	fantasy	novels	written	by	British	author	J.	K.	Rowling
Basic-level Masking	[mask]	Potter	is	a	series	[mask]	fantasy	novels	[mask]	by	British	author	J.	[mask]	Rowling
Entity-level Masking	Harry	Potter	is	a	series	[mask]	fantasy	novels	[mask]	by	British	author	[mask]	[mask]	[mask]
Phrase-level Masking	Harry	Potter	is	[mask]	[mask]	[mask]	fantasy	novels	[mask]	by	British	author	[mask]	[mask]	[mask]

Figure 2: Different masking level of a sentence

Experiment

4.1 Heterogeneous Corpus Pre-training

mixed corpus Chinese Wikepedia, Baidu Baike, Baidu news and Baidu Tieba.

The number of sentences are 21M, 51M, 47M, 54M.

Baidu Baike: encyclopedia articles, a strong basis for language modeling.

Baidu news: latest information about movie names, actor names, football team names, etc.

Baidu Tieba: open discussion forum, can be regarded as a dialogue thread.

4.2 DLM (Dialogue Language Model)

Dialogue data is important for semantic representation

ERNIE models the Query-Response dialogue structure on the DLM (Dialogue Language Model) task.

our method introduces dialogue embedding to identify the roles in the dialogue

The DLM task helps ERNIE to learn the implicit relationship in dialogues

4.2 DLM

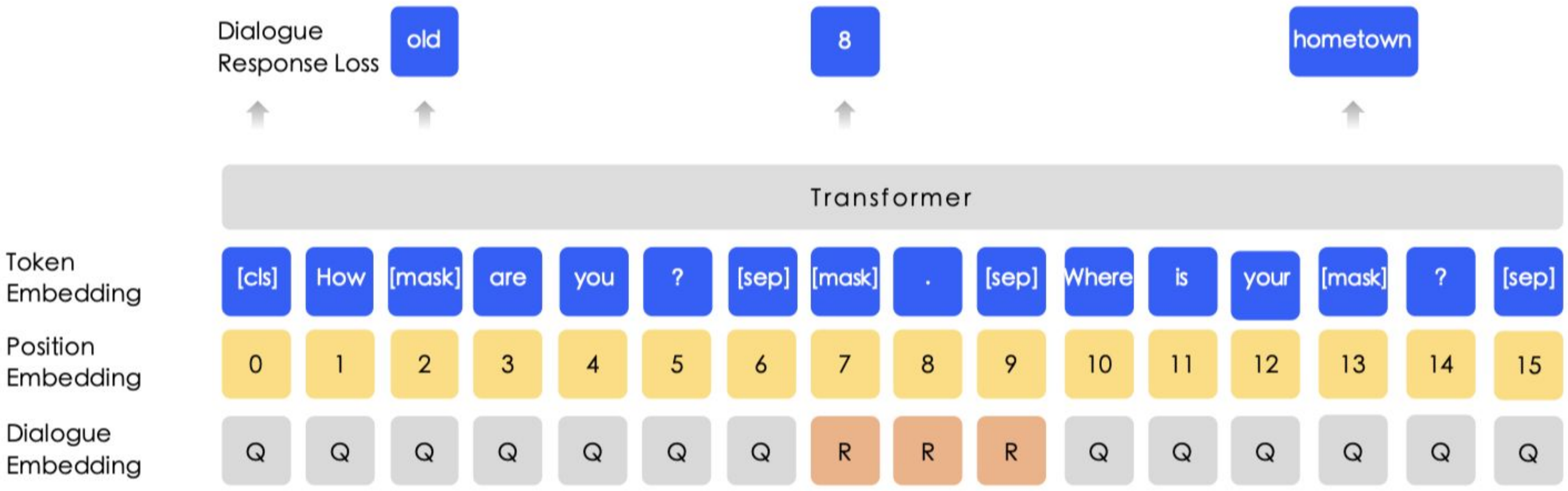


Figure 3: Dialogue Language Model. Source sentence: [cls] How [mask] are you [sep] 8 . [sep] Where is your [mask] ? [sep]. Target sentence (words the predict): old, 8, hometown)

4.3 Experiments on Chinese NLP Tasks

4.3.1 Natural Language Inference
4.3.2 Semantic Similarity
4.3.3 Name Entity Recognition
4.3.4 Sentiment Analysis
4.3.5 Retrieval Question Answering

4.4 Experiment results

Table 1: Results on 5 major Chinese NLP tasks

Task	Metrics	Bert		ERNIE	
		dev	test	dev	test
XNLI	accuracy	78.1	77.2	79.9 (+1.8)	78.4 (+1.2)
LCQMC	accuracy	88.8	87.0	89.7 (+0.9)	87.4 (+0.4)
MSRA-NER	F1	94.0	92.6	95.0 (+1.0)	93.8 (+1.2)
ChnSentiCorp	accuracy	94.6	94.3	95.2 (+0.6)	95.4 (+1.1)
nlpcc-dbqa	mrr	94.7	94.6	95.0 (+0.3)	95.1 (+0.5)
	F1	80.7	80.8	82.3 (+1.6)	82.7 (+1.9)

4.5 Ablation Studies

4.5.1 Effect of Knowledge Masking Strategies

Table 2: XNLI performance with different masking strategy and dataset size

pre-train dataset size	mask strategy	dev Accuracy	test Accuracy
10% of all	word-level(chinese character)	77.7%	76.8%
10% of all	word-level&phrase-level	78.3%	77.3%
10% of all	word-level&phrase-leve&entity-level	78.7%	77.6%
all	word-level&phrase-level&entity-level	79.9 %	78.4%

4.5 Ablation Studies

4.5.2 Effect of DLM

Table 3: XNLI finetuning performance with DLM

corpus proportion(10% of all training data)	dev Accuracy	test Accuracy
Baike(100%)	76.5%	75.9%
Baike(84%) / news(16%)	77.0%	75.8%
Baike(71.2%)/ news(13%)/ forum Dialogue(15.7%)	77.7%	76.8%

4.6 Cloze Test

No	Text	Predict by ERNIE	Predict by BERT	Answer
1	2006年9月，______与张柏芝结婚，两人婚后育有两儿子——大儿子Lucas谢振轩，小儿子Quintus谢振南；	谢霆锋	谢振轩	谢霆锋
	In September 2006, _____ married Cecilia Cheung. They had two sons, the older one is Zhenxuan Xie and the younger one is Zhennan Xie.	Tingfeng Xie	Zhenxuan Xie	Tingfeng Xie
2	戊戌变法，又称百日维新，是______、梁启超等维新派人士通过光绪帝进行的一场资产阶级改良。	康有为	孙世昌	康有为
	The Reform Movement of 1898, also known as the Hundred-Day Reform, was a bourgeois reform carried out by the reformists such as ____ and Qichao Liang through Emperor Guangxu.	Youwei Kang	Shichang Sun	Youwei Kang
3	高血糖则是由于______分泌缺陷或其生物作用受损，或两者兼有引起。糖尿病时长期存在的高血糖，导致各种组织，特别是眼、肾、心脏、血管、神经的慢性损害、功能障碍。	胰岛素	糖糖内	胰岛素
	Hyperglycemia is caused by defective ______ secretion or impaired biological function, or both. Long-term hyperglycemia in diabetes leads to chronic damage and dysfunction of various tissues, especially eyes, kidneys, heart, blood vessels and nerves.	Insulin	(Not a word in Chinese)	Insulin
4	澳大利亚是一个高度发达的资本主义国家，首都为______。作为南半球经济最发达的国家和全球第12大经济体、全球第四大农产品出口国，其也是多种矿产出口量全球第一的国家。	墨尔本	墨悉本	堪培拉
	Australia is a highly developed capitalist country with _______ as its capital. As the most developed country in the Southern Hemisphere, the 12th largest economy in the world and the fourth largest exporter of agricultural products in the world, it is also the world's largest exporter of various minerals.	Melbourne	(Not a city name)	Canberra (the capital of Australia)
5	______是中国神魔小说的经典之作，达到了古代长篇浪漫主义小说的巅峰，与《三国演义》《水浒传》《红楼梦》并称为中国古典四大名著。	西游记	《小》	西游记
	__________is a classic novel of Chinese gods and demons, which reaching the peak of ancient Romantic novels. It is also known as the four classical works of China with Romance of the Three Kingdoms, Water Margin and Dream of Red Mansions.	The Journey to the West	(Not a word in Chinese)	The Journey to the West
6	相对论是关于时空和引力的理论，主要由______创立。	爱因斯坦	卡尔斯所	爱因斯坦
	Relativity is a theory about space-time and gravity, which was founded by _________.	Einstein	(Not a word in Chinese)	Einstein

Figure 4: Cloze test

Conclusion

Conclusion

we presents a novel method to integrate knowledge into pre-training language model.

our method outperforms BERT over all of these tasks.

both the knowledge integration and pre-training on heterogeneous data enable the model to obtain better language representation.